

Analytic problem
solving

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Slot C

1.1. Grey level Transformation

a. Purpose of Grey level Transformation

→ Grey level transformation is used to modify the intensity values of an image to improve visual appearance. It enhances brightness and contrast, highlights important features, and suppresses unwanted details. These transformation are commonly used in image processing before segmentation and object recognition.

b. Negative Transformation.

Formula

$$s = L - 1 - r$$

where,

$L = 256$ (for an 8-bit image)

r = original pixel value

s = transformed pixel value

~~2. Histogram Equalization~~

~~When contrast is low
and intensity values
are concentrated in a small range~~

~~then image contrast is low
and details are hidden.~~

zed intense
do:

$$S = \frac{C_o}{g_b}$$

Intensity

Equalized

Interpretation

a. Histogram Equalization

a. Need for Histogram Equalization

Histogram equalization improves image contrast by redistribution pixel intensity values over the available range. It enhances availability of visibility of details in low-contrast images.

b. CDF calculation:

Histogram:

Intensity	Frequency	CDF
0	4	4
1	6	10
2	10	20
3	6	26

$\therefore$ Total pixels = 26

Equalized Intensities

Formula

$$S = \frac{DF}{255} \times 255$$

Intensity	Equalized value
0	0
1	1
2	2
3	3

Interpretation: Histogram equalization spreads intensity values more uniformly, producing a clearer image with better contrast.

Gaussian Smoothing

a. Why Gaussian Filtering:

Removes random noise

Preserves image edges better than averaging filters.

Gives more importance to nearby pixels.

Produces smooth and natural-looking images.

Widely used before edge detection.

b. Calculation:

The Gaussian kernel and image values are ~~as follows~~

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$\begin{aligned}
 &= (1 \times 10) + (2 \times 20) + (1 \times 10) \\
 &+ (2 \times 20) + (1 \times 10) + (2 \times 20) + (1 \times 10) \\
 &= 10 + 40 + 10 + 40 + 10 + 40 + 10 + 10
 \end{aligned}$$

$$= \frac{360}{16}$$

22.5

Noise is reduced at 22.5
Image becomes smoother

④ Edge Detection using Sobel operator:

or Purpose:

Edge detection identifies boundaries between objects by detecting sudden changes in pixel intensity. It is used in image segmentation, feature extraction, and object recognition.

or Calculation:

The sobel mask

$$= \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

the image matrix

$$= \begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$$= (-1 \times 10) + (-2 \times 10) + (1 \times 10)$$

$$= -10 - 20 + 10$$

$$= -40$$

$$= (3 \times 30) + (2 \times 30) + (1 \times 30)$$

$$= 30 + 60 + 30$$

$$= 120$$

$$G = -40 + 120$$

$$= 80$$

Interpretation:

A strong horizontal edge is detected.

Large gradient value indicate stronger edges.

Contrast Stretching:

a. Need for contrast stretching

Contrast stretching increases the dynamic range of image intensities. It improves the visibility of objects.

Formula:

$$s = \frac{(r - 50) \times 255}{150 - 50}$$

$$r = 50 \quad s = \frac{50 - 50}{150 - 50} \times 255$$

$$\boxed{s = 0}$$

$$x = 100$$

$$S = \frac{100 - 50}{100} \times 115$$

$$[S = 100]$$

$$x = 150$$

$$\Rightarrow S = \frac{150 - 50}{100} \times 115$$

$$[S = 255]$$

original



50

100

150

Transformed

0

128

255